

Girsanov (in finite time)

Let $\gamma \in \mathcal{L}$ such that $\mathcal{E}(\gamma \bullet W)$ is a (true) martingale on $[0, T^*]$ under $\mathbb{P}$. Then

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \mathcal{E}_T(\gamma \bullet W) \tag{1}$$

Defines an equivalent probability measure $\mathbb{Q} \sim \mathbb{P}$ and

$$W^*(t) = W(t) - \int_0^t \gamma(s)^\top ds$$

defines a brownian motion under $\mathbb{Q}$

Girsanov (in finite time)

Let $\gamma^* \in \mathcal{L}$ such that $\mathcal{E}(\gamma^* \bullet W^*)$ is a (true) martingale on $[0, T^*]$ under $\mathbb{Q}$. Then

$$\frac{d\mathbb{N}}{d\mathbb{Q}} = \mathcal{E}_T(\gamma^* \bullet W^*) \tag{2}$$

Defines an equivalent probability measure $\mathbb{N} \sim \mathbb{P}$ and

$$W^{\mathbb{N}}(t) = W^*(t) - \int_0^t \gamma^*(s)^\top ds$$

defines a Brownian motion under $\mathbb{N}$

We want to define a new measure, the forward measure $\mathbb{Q}^T$, which is given by:

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} = \mathcal{E}_T(\nu(\cdot, T) \cdot W^*)$$

with $\nu(t, T) = \int_t^T \sigma^T(t, u) du$ taken from P-dynamics

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_t} = \frac{P(t, S)P(0, T)}{P(t, T)P(0, S)} = \mathcal{E}_t((\nu(\cdot, S) - \nu(\cdot, T)) \cdot W^T) = \mathcal{E}_t(\sigma_{t,T}^* \bullet W^T)$$

$P \sim \mathbb{Q} : \mathbb{E}^{\mathbb{Q}}[X| \mathcal{F}] = \frac{\mathbb{E}^{\mathbb{P}}[X \frac{d\mathbb{Q}}{d\mathbb{P}} | \mathcal{F}]}{\mathbb{E}^{\mathbb{P}}[\frac{d\mathbb{Q}}{d\mathbb{P}} | \mathcal{F}]}, \quad \mathbb{E}^{\mathbb{P}}[X| \mathcal{F}] = \frac{\mathbb{E}^{\mathbb{Q}}[X \frac{d\mathbb{P}}{d\mathbb{Q}} | \mathcal{F}]}{\mathbb{E}^{\mathbb{Q}}[\frac{d\mathbb{P}}{d\mathbb{Q}} | \mathcal{F}]}$

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} \Big|_{\mathcal{F}_t} = \frac{P(t, S)P(0, T)}{P(t, T)P(0, S)} \quad \frac{d\mathbb{Q}}{d\mathbb{Q}^T} = \frac{B(t)P(0, T)}{P(t, T)} \quad \frac{d\mathbb{Q}^T}{d\mathbb{Q}} = \frac{1}{P(0, T)B(t)} \quad \frac{d\mathbb{Q}}{d\mathbb{Q}^T} \Big|_{\mathcal{F}_t} = \frac{B(t)P(0, S)}{P(t, S)}$$

why? Because its a martingale. Important!

Essentially what were doing is defining a measure, where $P(t, T)$ acts as the numeraire.

• Under $\mathbb{Q}^T$ then $\frac{P(t, S)}{P(t, T)}$ is a martingale.

• Under $\mathbb{Q}$ then $\frac{P(t, T)}{B(t)}$ is a martingale.

To see why this is better consider that we for a T-claim X can find:

$$B(t) \mathbb{E}^{\mathbb{Q}} \left[\frac{X}{B(T)} \right] = P(t, T) \mathbb{E}_t^{\mathbb{Q}^T} [X] \quad (\text{Proof easy})$$

Stochastic interest rates. Means we have to consider a joint distribution of B and X to determine expectation. Much easier to calculate/simulate

Recall Q-dynamics of $f(t, T)$:

$$df(t, T) = \sigma(t, T) \left(\int_t^T \sigma(t, u)^\top du \right) dt + \sigma(t, T) dW^*(t)$$

Use Girsanov to switch measures:

$$dW^*(t) = dW^\top + \nu(t, T)^\top = dW^\top - \int_t^T \sigma(t, u)^\top du \, dt$$

Inserting:

$$df(t, T) = \sigma(t, T) dW^\top$$

Thus $df(t, T)$ is a martingale under $\mathbb{Q}^T$, from which it follows:

$$f(t, T) = \mathbb{E}_t^{\mathbb{Q}^T} [f(T, T)] = \mathbb{E}^{\mathbb{Q}^T} [r(-t)]$$

Thus under $\mathbb{Q}^T$ forward rates can be seen as forecasts of future rates.

Example: Caplet

A LIBOR caplet pays $S(F(T, T+\delta) - K)^+$ at time $T+\delta$. Single curve and skipping some calculations:

$$\begin{aligned} C_P(0) &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^{T+\delta} r(s) ds} (S(F(T, T+\delta) - K))^+ \right] \\ &\stackrel{\text{BS}}{\approx} \frac{1}{K} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^{T+\delta} r(s) ds} (K - P(T, T+\delta)) \mathbb{1}_{\{K > P(T, T+\delta)\}} \right], \quad K \approx T + \delta R \\ &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^{T+\delta} r(s) ds} \mathbb{1}_{\{K > P(T, T+\delta)\}} \right] - \frac{1}{K} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^{T+\delta} r(s) ds} P(T, T+\delta) \mathbb{1}_{\{K > P(T, T+\delta)\}} \right] \\ &= P(0, T) \mathbb{E}^{\mathbb{Q}^T} [\mathbb{1}_{\{1 > \frac{1}{K} \}}] - \frac{1}{K} P(0, T+\delta) \mathbb{E}^{\mathbb{Q}^T} \left[\frac{P(T, T+\delta)}{P(T, T+\delta)} \mathbb{1}_{\{1 > \frac{1}{K} \}} \right] \\ &= P(0, T) \mathbb{Q}^T(R > P(T, T+\delta)) - \frac{P(0, T+\delta)}{K} \mathbb{Q}^{T+\delta}(K > P(T, T+\delta)) \end{aligned}$$

If we assume $P(t, T)$ is log-normal under both $\mathbb{Q}^T$ and $\mathbb{Q}^{T+\delta}$, then we can get a BS-like formula.

$$\mathbb{E}^{\mathbb{Q}^T} \left[\frac{B(t)}{P(t, T)} \right] = \frac{\mathbb{E}^{\mathbb{Q}} \left[\frac{B(t)}{P(t, T)} \frac{1}{P(0, T) B(T)} \right]}{\mathbb{E}^{\mathbb{Q}}}$$

Affinity under a forward measure

For $S \geq T \geq t$ we have

$$\mathbb{E}_t^{\mathbb{Q}^S} \left[e^{\int_t^T (\nu_0 + \nu_2 X_s) ds + w \cdot X_T} \right] = e^{\phi^S(t, T, \nu_0, \nu_2, w) + \psi^S(t, T, \nu_2, w) \cdot X_t}$$

For deterministic functions with terminal conditions $\phi^S(T, T, \nu_0, \nu_2, w) = 0$ and $\psi^S(T, T, \nu_2, w) = w$, given as

$$\begin{aligned} \phi^S(t, T, \nu_0, \nu_2, w) &= -\phi(t, S, -\rho_0, -\rho_1, 0) + \phi(T, S, -\rho_0, -\rho_1, 0) & (40) \\ &+ \phi(t, T, \nu_0 - \rho_0, \nu_1 - \rho_1, w + \psi(T, S, -\rho_1, 0)) & (41) \\ \psi^S(t, T, \nu_2, w) &= -\psi(t, S, -\rho_1, 0) + \psi(t, T, \nu_2 - \rho_1, w + \psi(T, S, -\rho_1, 0)) & (42) \end{aligned}$$

Let X be an arbitrary Affine Process.

Affine Definitions

$$\mathbb{E}_t^{\mathbb{Q}} \left[e^{\int_t^T (\nu_0 + \nu_2 X_s) ds + w \cdot X_T} \right] = e^{\psi(t, T, \nu_0, \nu_2, w) + \psi(t, T, \nu_2, w) \cdot X_t} \tag{37}$$

$$R(t) = \rho_0 + \rho_1 \cdot X_t, \quad B(t) = e^{\int_t^T R(s) ds} \tag{38}$$

$$P(t, T) = \mathbb{E}_t^{\mathbb{Q}} \left[e^{-\int_t^T (\nu_0 + \nu_2 X_s) ds} \right] = e^{\phi(t, T, -\rho_0, -\rho_1, 0) + \psi(t, T, -\rho_1, 0) \cdot X_t} \tag{39}$$

With ϕ and ψ defined in the usual way but now written explicitly as a function of ν_0, ν_1 and w

Assuming deterministic $\sigma(t, T)$ results in log-normally distributed zero-coupon bond prices:

$$\begin{aligned} \frac{P(T, T+\delta)}{P(t, T)} &= \frac{P(0, T+\delta)}{P(0, T)} \mathcal{E}(\sigma_{t, T+\delta} \bullet W^T) \quad (\text{Lemma 7.1, slide 5}) \\ &= \frac{P(0, T+\delta)}{P(0, T)} \exp \left(\int_0^T \sigma_{t, T+\delta}(s) dW^T(s) - \frac{1}{2} \int_0^T \|\sigma_{t, T+\delta}(s)\|^2 ds \right) \sim e^{\tilde{X}}, \quad X \sim \mathcal{N} \end{aligned}$$

Which means the forward rate cannot be log-normal:

$$F(T, T+\delta) = \frac{1}{\delta} \left(\frac{1}{P(T, T+\delta)} - 1 \right)$$

How to unify with real-world? Market practitioners heavily relied on Black-76 - a model that assumes the forward rate is log-normal.

Known as LIBOR Market Model. We can also define LMM without any reference to HJM. Pro: Easier. Cons: Model produces only a discrete term structure.

The given formula is without HJM. Allows us to model Caplet-prices like BS-76.
 $dL(t, T) = L(t, T) \lambda(t, T) dW^{T, \lambda}(t)$
 $\lambda(t, T) = \frac{1}{P(t, T+\delta)} - 1$
 to price Caplets

Lemma 11.1

The dynamics of $L(t, T_n)$ under $\mathbb{Q}^{T, n+1}$ for $t \in (0, \min(T_n, T_n))$

$$\frac{dL(t, T_n)}{L(t, T_n)} = \begin{cases} -\lambda(t, T_n) \sum_{i=m+1}^n \lambda(t, T_i)^\top \frac{\partial L(t, T_i)}{\partial L(t, T_i)} dt + \lambda(t, T_n) dW^{T, n+1} & m < n \\ \lambda(t, T_n) dW^{T, n+1} & m = n \\ +\lambda(t, T_n) \sum_{i=m+1}^n \lambda(t, T_i)^\top \frac{\partial L(t, T_i)}{\partial L(t, T_i)} dt + \lambda(t, T_n) dW^{T, n+1} & m > n \end{cases} \tag{93}$$

Thus we can define a measure $\mathbb{Q}^{T, n+1}$ for each rate $L(t, T_n)$, where the rate is log-normal (n.n.m). Problem: Sometimes we want to price multi-rate products (like swaptions, whose payoffs depend on a whole basket of rates. Here Monte Carlo is our only option: Essentially just fix a single measure and then define dynamics (state dependence in other measures is hard in closed form). Using MC on Terminal measure tends to make the error explode. Fix: Use spot rate measure, which essentially just has the numeraire B^0 which is a numeraire that grows at LIBOR rate.

Lemma 11.6

$$\frac{dL(t, T_n)}{L(t, T_n)} = \lambda(t, T_n) \sum_{i=m+1}^n \lambda(t, T_i)^\top \frac{\partial L(t, T_i)}{\partial L(t, T_i)} dt + \lambda(t, T_n) dW^{\mathbb{Q}^T}(t) \tag{117}$$

Negative interest rates: the log-normal model doesn't allow for negative rates which is problematic. Say we have a negative strike K , such that a caplet is priced:
 $(L - K)^+ \xrightarrow[\text{negative}]{\text{always positive}} \text{always greater than 0.}$

Just add a positive constant α to the rate so the new log-normal variable $K(t) = L(t, T) + \alpha$ can allow for negative $L(t, T)$ -values. Careful with this method! Produces an asymmetric vol smile.

Lemma 11.1 Displaced Diffusion

The dynamics of $L(t, T_n)$ under $\mathbb{Q}^{T, n+1}$ for $t \in (0, \min(T_n, T_n))$

$$\frac{dL(t, T_n)}{L(t, T_n) + \alpha} = \begin{cases} -\lambda(t, T_n) \sum_{i=m+1}^n \lambda(t, T_i)^\top \frac{\partial (L(t, T_i) + \alpha)}{\partial (L(t, T_i) + \alpha)} dt + \lambda(t, T_n) dW^{T, n+1} & m < n \\ \lambda(t, T_n) dW^{T, n+1} & m = n \\ +\lambda(t, T_n) \sum_{i=m+1}^n \lambda(t, T_i)^\top \frac{\partial (L(t, T_i) + \alpha)}{\partial (L(t, T_i) + \alpha)} dt + \lambda(t, T_n) dW^{T, n+1} & m > n \end{cases} \tag{94}$$

And the construction of the displaced diffusion LMM follows as before.

Alternatively, use Bachelier LMM:

$$dL(t, T_{n-1}) = \lambda(t, T_{n-1}) dW^{T, n}(t)$$

No more state dependence in volatility $\rightarrow$ Rate doesn't get stuck at small values.

Slide 10: The Swap Measure

What to show on the slide:

- The Instrument: Swaption (An option to enter an interest rate swap).
- The Numeraire: The Annuity, $A(t) = \sum_{i=m+1}^n \delta P(t, T_i)$
- The Result: The forward swap rate $R_{\text{swap}}(t)$ is a martingale under the Swap Measure $\mathbb{Q}^{\text{swap}}$.

What to say (The Script):

"To wrap things up, let's look at one of the most heavily traded derivatives in the market: the swaption. A swaption is simply an option that gives you the right to enter into an interest rate swap at a future date. If we want to price a swaption cleanly, the standard forward measures we've been using are actually a bit clumsy. Instead, we introduce a new numeraire called the 'Annuity' or 'Present Value of a Basis Point', denoted as $A(t)$. The Annuity is just a portfolio of zero-coupon bonds corresponding to the payment dates of the swap. Because it's a portfolio of tradable assets, it is a perfectly valid numeraire. When we change our measure using this Annuity as our numeraire, we move into what is called the 'Swap Measure'. The beauty of this is that under the Swap Measure, the forward swap rate $R_{\text{swap}}(t)$ becomes a pure martingale. This allows us to price swaptions using a clean, Black-Scholes-style closed-form solution, effectively treating the swap rate itself as the underlying asset."

Slide 11: The Grand Contradiction & Conclusion

What to show on the slide:

- The Math Fact: The sum of log-normal variables is not log-normal.
- The Contradiction: LMM (Log-normal Forwards) vs. Swap Market Model (Log-normal Swap Rates).
- The Takeaway: Pick the model that fits the specific derivative book.

What to say (The Script):

"But this brings us to the grand contradiction of modern interest rate modeling. If you look closely at a swap rate, it is mathematically just a weighted average of individual forward rates. Here is the mathematical catch: the sum of log-normally distributed variables is not log-normal. This means we have a strict theoretical conflict. You cannot live in a universe where the LIBOR Market Model is true—meaning all discrete forward rates are log-normal—while the Swap Market Model is also true—meaning the swap rate is log-normal. Assuming both simultaneously introduces arbitrage into your math. So, how do practitioners survive this contradiction? By being deeply pragmatic. You don't ask which model is universally 'true'; you ask which model is appropriate for the risk you are managing. If you are a trader pricing a single, plain-vanilla European swaption, you use the Swap Market Model because it cleanly targets that local risk. But, if you are a quant managing a massive, complex book of exotic, path-dependent derivatives that rely on the entire term structure, you accept the LMM. You sacrifice the perfect closed-form swaption formula in exchange for a globally consistent, arbitrage-free model of the entire yield curve. In the end, theoretical purity gives way to practical utility. Thank you."